

ACADEMIC TRACK: DISTRIBUTED SYSTEMS & CLOUD ARCHITECTURE

Analysis of Raft Consensus Protocol Under Asymmetric Network Partitions

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Abstract: Consensus protocols serve as the foundational backbone for resilient state machine replication in distributed datastores. In this paper, we present an empirical fault-injection evaluation of the Raft consensus protocol operating under asymmetric network partitions and packet drop scenarios. We demonstrate how partitioned leader isolation affects log compaction and examine quorum election safety under bounded network delays. Experimental benchmarks across a 7-node cluster show zero data loss with median leader transition latencies below 142 ms.

1. Introduction and Problem Formulation

Modern distributed databases require fault tolerance guarantees under crash-fault assumptions. While Paxos provides theoretical optimality, its operational complexity often leads to flawed implementations. Raft decouples consensus into distinct subproblems: leader election, log replication, and commitment safety.

2. System Architecture & Quorum Replication Protocol

Our experimental cluster comprises 7 replicated state machines executing synchronized log entry compaction. A term counter t monotonically increments upon every election cycle. Commit safety is verified when at least 4 of 7 nodes acknowledge the entry.

Cluster Size	Consensus Protocol	Avg Latency	Peak QPS	Failover Latency	Safety Proof
3 Nodes	Raft (Standard)	4.2 ms	14,800 ops/s	118 ms	Verified
5 Nodes	Raft (Optimized)	5.8 ms	13,200 ops/s	135 ms	Verified
7 Nodes	Raft (Partitioned)	7.4 ms	11,400 ops/s	142 ms	Verified
7 Nodes	Multi-Paxos	6.9 ms	11,800 ops/s	190 ms	Verified

3. Fault-Tolerance & Partition Recovery Evaluation

During minute 4 of continuous load testing, the primary leader node was isolated within an asymmetric minority partition. Within 138 ms, follower nodes detected heartbeat timeouts and elected a new leader without split-brain divergence.

4. Conclusion & Academic Rigor

The architecture achieves linearizable read guarantees and rapid failover. Raft sustains 11,400 ops/sec throughput under active minority node failures.